Plan B: The Show Must Go On

Deciding what to verify, what to prepare, and when to commit a change

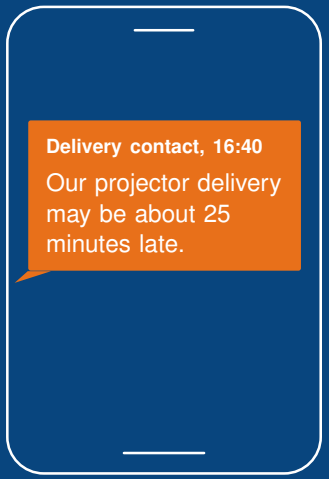
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University of Massachusetts Boston

Paul English Applied AI Institute (PEAAII) Hackathon, UMass Boston, October 2026



16:40. Twenty minutes to doors open.



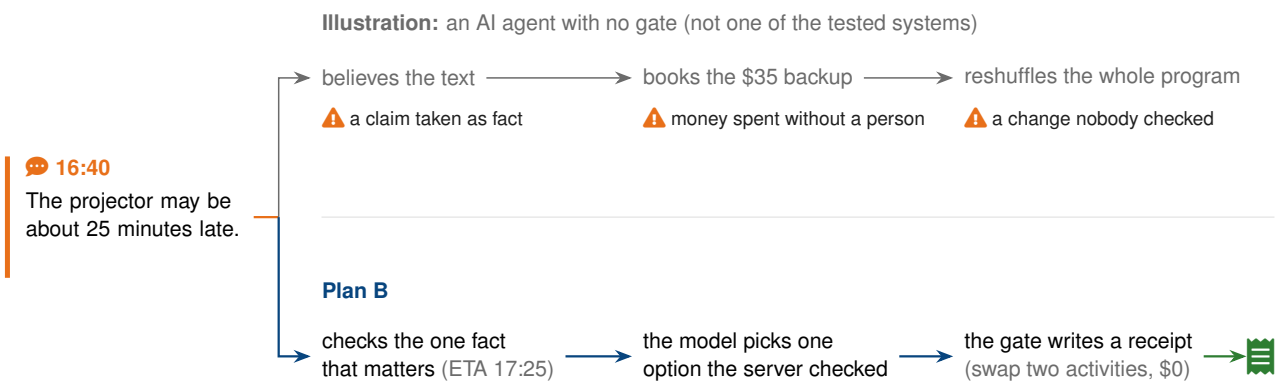


20 minutes to doors open

Guests arrive at 17:00. The demonstrations need the projector.

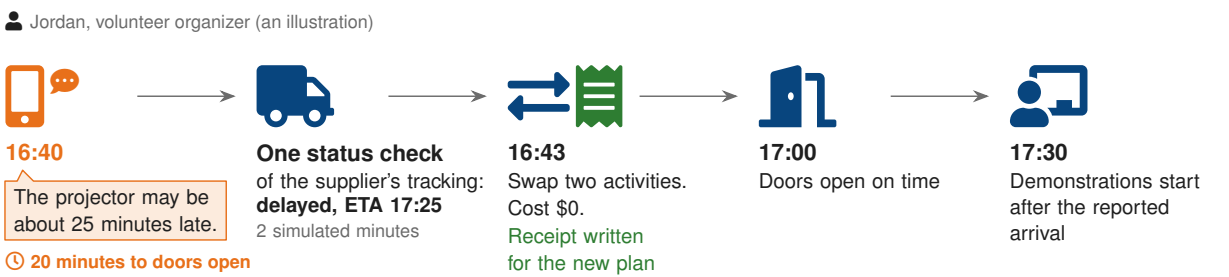
Jordan, a volunteer organizer, is an illustration, and the campus showcase is a simulated event in a local sandbox.

An agent that acts on a text can spend money on a rumor, so Plan B checks first



In Plan B the model never acts on its own. It picks one option the server checked, and **one gate** writes the receipt.

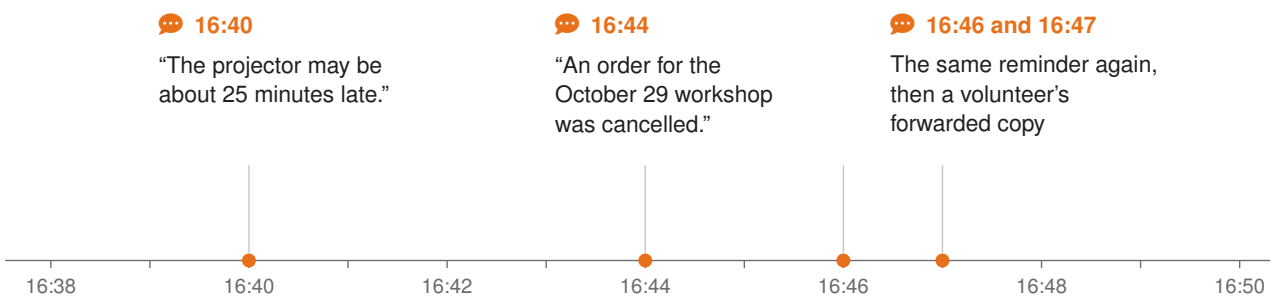
Plan B checked one fact, swapped two activities, and the doors opened on time



The planner built **16 options** and **3 passed** every check. The swap cost $0.

A development run in the sandbox with the fixed rules. In a recorded run, our coordinator made the same check and the same swap at 16:43.

Most messages should change nothing, so Plan B checks only what could change the decision

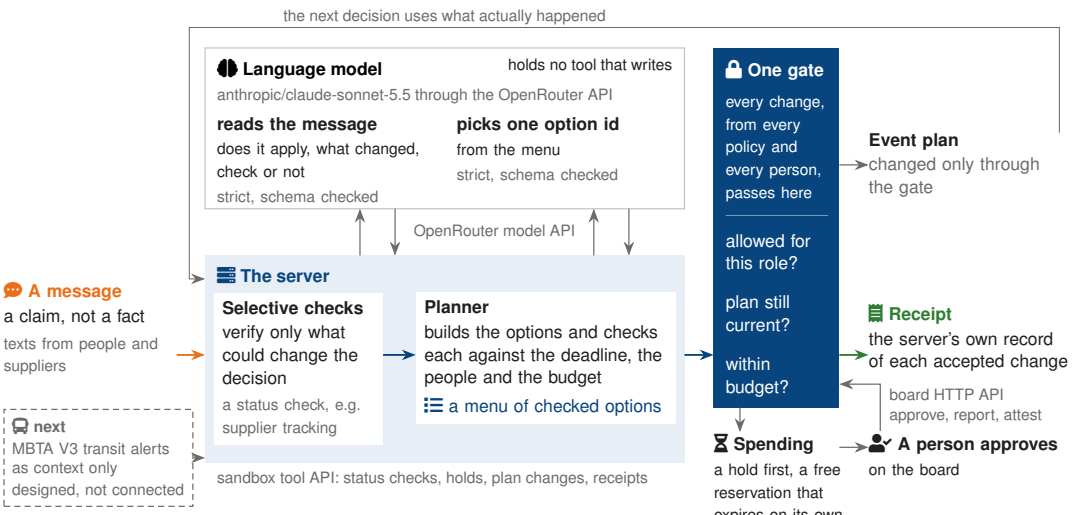


✔ **The plan is kept: no hold, no spending, no change.**

On the new test cases, our coordinator made 0.8 status checks per campus run and the check-all variant 1.5, with the same outcomes.

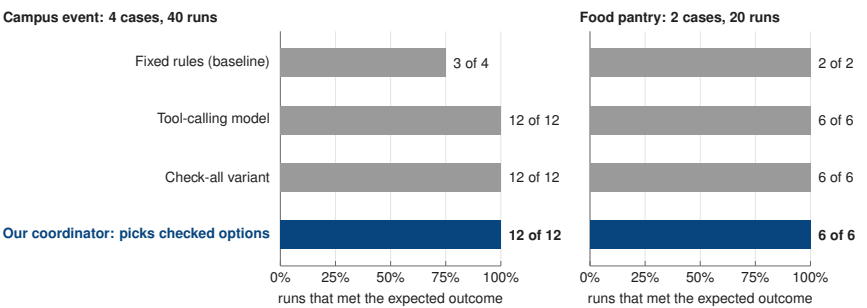
A second development case, in the order it happened, decided by the fixed rules in an automated run.

THE MODEL ONLY PICKS AMONG OPTIONS THE SERVER BUILT AND CHECKED, AND ONE GATE WRITES EVERY RECEIPT



The model: Claude Sonnet 5.5 (`anthropic/claude-sonnet-5.5`) through the OpenRouter API, the same model and settings for all three model approaches. Only how the model is used differs. Spending starts as a **hold**, a free reservation that expires on its own, until a person approves.

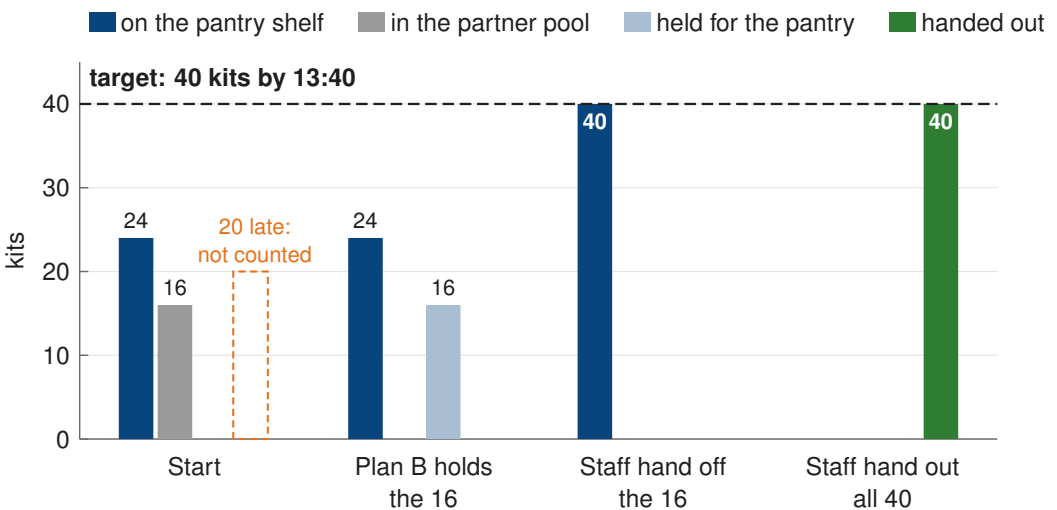
On 6 new cases, 59 of 60 runs met the expected outcome, and the one miss was the fixed rules



The runs: each new case ran 10 times, the fixed rules once (no model) and each model approach 3 times, all on Claude Sonnet 5.5 through OpenRouter. 4 campus cases make 40 runs, 2 pantry cases make 20. **No executed action broke a rule**: the gate refused 6 attempts and recorded each.

The six new cases. Campus: a real projector delay, a believable false alarm, a volunteer who runs late and then checks in, and a disruption with no allowed fix. Pantry: a late shipment while a partner can lend kits, and a false “it arrived” claim. Written and locked before the first model call. An earlier 12-case test found real defects, fixed before the freeze.

THE SAME LOOP RUNS A FOOD PANTRY, WHERE FOOD COUNTS ONLY ONCE IT IS ON THE SHELF



Plan B can hold the partner's 16 kits, but **only staff** record a handoff or a distribution.

A development case of the fictional pantry sandbox. The two new pantry cases are in the test on the previous slide.

Next: real feeds as context, and a pilot where Plan B proposes and the organizer acts

Connected today

OpenRouter model API
anthropic/claude-sonnet-5.5 reads messages and picks options

The sandbox's tool API
status checks, holds, plan changes, receipts

The board's HTTP API
people approve spending, report and attest

Next

MBTA V3 API, live transit alerts
designed, not connected: context only, never a person's arrival

Real tracking pages
a supplier's own status in place of the sandbox's

A pilot with a campus event series
Plan B proposes, the organizer acts. Count messages, checks, changes and approvals

A late train could only ever be context. It never proves that a person is late.

Everything shown runs in a local sandbox. No real bookings, stock or partners, and no community has used it yet.

16:40 AGAIN. CHECK FIRST, THEN CHANGE.

Plan B helps a community event stay on track by deciding what to verify, what to prepare, and when to commit a change.

At 17:00 the doors opened on time: one check, a free swap, and a receipt.

Noctis Zhang, University of Massachusetts Boston

UMass Boston

github.com/nocdoggo/FA26-COLLIE

Backup: the six new cases and the sixty runs, one by one

New case	What happens	The right outcome	Fixed rules	Each model approach
Campus, real delay	The projector delivery will miss its slot	Move the demonstrations ten minutes and finish on time	met, 1 of 1	met, 3 of 3
Campus, false alarm	A relayed call says the delivery is late, the order desk says on schedule	Change nothing	met, 1 of 1	met, 3 of 3
Campus, late volunteer	The leader of the first activity reports a late bus, then checks in	Adjust from the check-in and still finish	**missed**, 0 of 1	met, 3 of 3
Campus, no allowed fix	The projector order is withdrawn and the spare needs confirming too far ahead	Report that no allowed fix exists and change nothing invalid	met, 1 of 1	met, 3 of 3
Pantry, late shipment	The main shipment arrives after service, a partner can lend 16 kits	Hold and receive the 16, reach the 40-kit target	met, 1 of 1	met, 3 of 3
Pantry, false claim	A sales rep says “delivered”, the real shipment comes after the deadline	Serve the 24 kits on hand and report the 16-kit gap	met, 1 of 1	met, 3 of 3

Four approaches. Fixed rules: written rules, no model. Tool-calling model: the model calls the sandbox tools itself. Check-all variant: the server runs every relevant status check first, then the model picks. Our coordinator: the model picks one option from the menu the server built and checked. The three model approaches all use Claude Sonnet 5.5 through OpenRouter.

BACKUP: WHAT THE NEW TEST COST, PER RUN

	Campus event		Food pantry	
Approach	**Model calls**	**USD per run**	**Model calls**	**USD per run**
Fixed rules (baseline)	0	0.00	0	0.00
Tool-calling model	36	0.10	38	0.19
Check-all variant	21	0.02	30	0.05
Our coordinator	30	0.03	39	0.07

Provider-reported cost. The whole new test cost $3.61 for 60 runs. Limits per run: 48 tool calls, 8 model calls, $0.50 and 900 seconds.

BACKUP: WHAT THESE NUMBERS DO NOT SHOW

Small and close to development. Six new cases, and all six repeat conditions seen in development, with similar wording. The case is the unit. Three runs of a case are repeats with the provider's default sampling, not independent evidence. No significance test.

The earlier locked test, before the fixes. Our coordinator met every run of 6 of 8 campus cases there, and the tool-calling model of 8 of 8. Those failures are what the fixes target. A rerun on those spent cases is a diagnostic, not a test and not comparable as an improvement: our coordinator meets every run of 7 of 8 campus cases there and still misses one case in all three runs. The documentation has the full record.

A gate by design. Zero executed violations is what the gate enforces. The refused attempts show it at work. Scoring is separate code that reads the final state and the receipts.

Backup: a partial handoff stays partial

In another development case of the pantry, staff hand off only **10 of the 16** kits. 34 kits are handed out and a 6-kit deficit is reported. The target stays 40.

Stock is conserved: 34 handed out plus 6 left in the partner pool make 40. The 20 kits of the late shipment never count as stock.

BACKUP: THE FROZEN CONFIGURATION

Model: anthropic/claude-sonnet-5.5 through the OpenRouter API, the same for the three model approaches, with strict JSON output.

Limits per run: 48 tool calls, 8 model calls, $0.50 and 900 seconds. Seeds 101, 202 and 303 are recorded but not sent, because the model's endpoints refuse a seed.

Frozen by hash before the first call on the new cases. The cases were locked seven seconds after the freeze, and the first model call came almost four minutes later.